

AMBIENT SYSTEMS

Climate At Your Service

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Cyber-Physical Systems (CPS) are driving the next wave of Enterprise Applications

Machine Intelligence

Edge Devices

Cyber Security

Automation



Custom Applications

Drivers

There is increasing demand for system optimization beyond energy efficiency.

Current Challenges:

1.Limited Resources
It's hard to find highly trained and experienced operators who can optimize multiple intelligent systems.

2. Technological Complexity
Many operate with outdated legacy systems and physical equipment that lack interoperability.

3. Competing Priorities
Different physical systems have different requirements based on the unique needs of occupants and operators.

Applied AI for Cyber-Physical Systems





Sense

IoT gateway and sensors co-designed with manufacturing partners

Optimize

NUS National University of Singapore

Patent Pending Invention from National University of Singapore (NUS)



Control

Offering end-to-end equipment controls after full product development

Technology

- Sensors are planted in different areas of the building for data collection.
- Different types of data (Air Quality, Equipment Settings, and User Preferences) are collected and displayed on our Dashboard for users to have an overview of their building's condition and performance.
- With Ambient System's machine learning, a set of instructions are recommended for users to adjust Equipment Settings (e.g. A.C. etc) to enhance building's performance.



Technology (How It Works)



Step 1: Connect

Integrate IoT with essential equipment and control systems

Step 2: Prioritize

All important data in one Dashboard

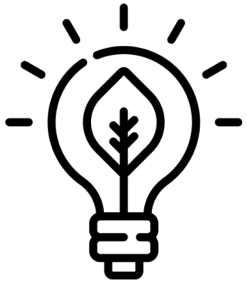
Step 3: Automate

Machine Learning-enabled optimization

Technology (Prioritize)

- Choose one of pre-programmed optimization plans or let the system diagnose areas for improvement
- Allocate weightage to each of the areas for improvement (e.g., 30%, 30%, 40%).

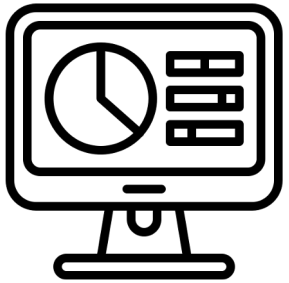
1. Energy Efficiency



2. Sustainability

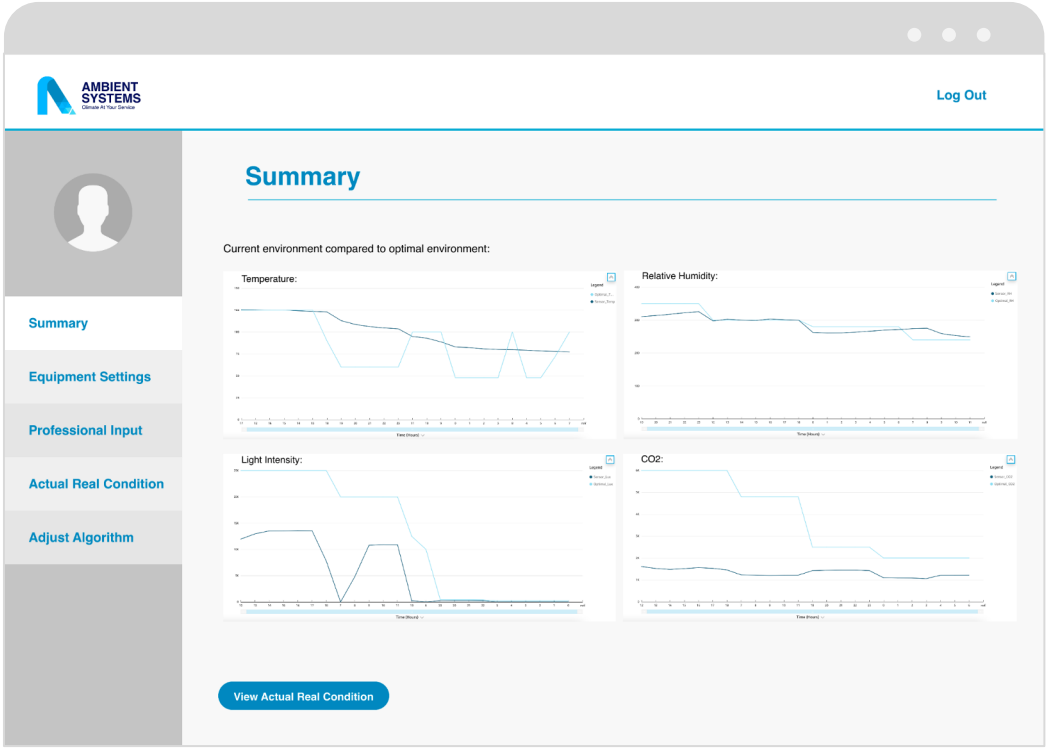


3. Output (Performance)



Technology (Dashboard)

Data collected from different parameters in one Dashboard



1. Dashboard

Overview of Building's data and performance.

2. Equipment Settings

Equipment settings set by users or daily operators.

3.

settings for building's best performance.

4. Actual Real Condition

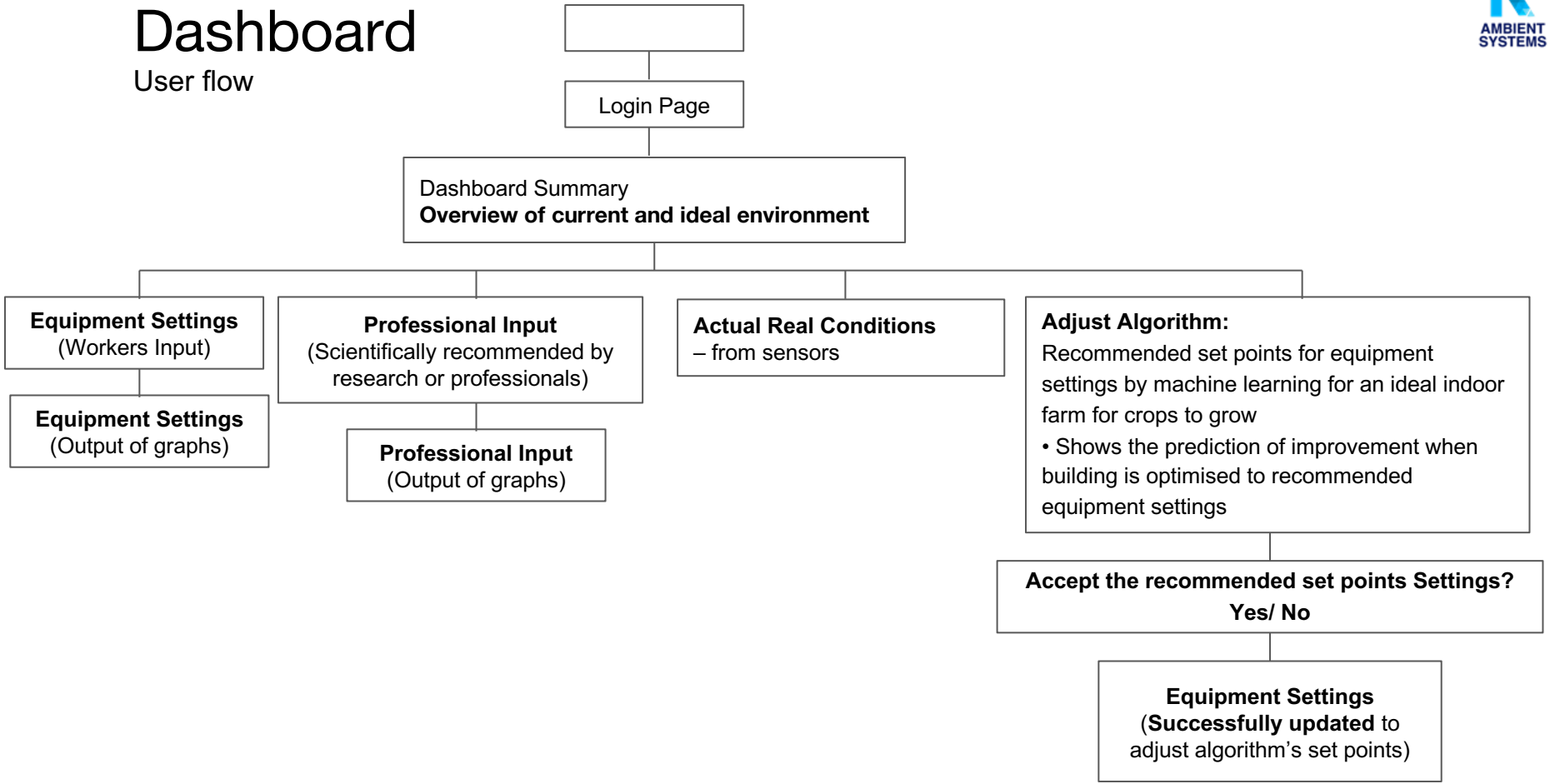
Data collected from different sensors.

5.

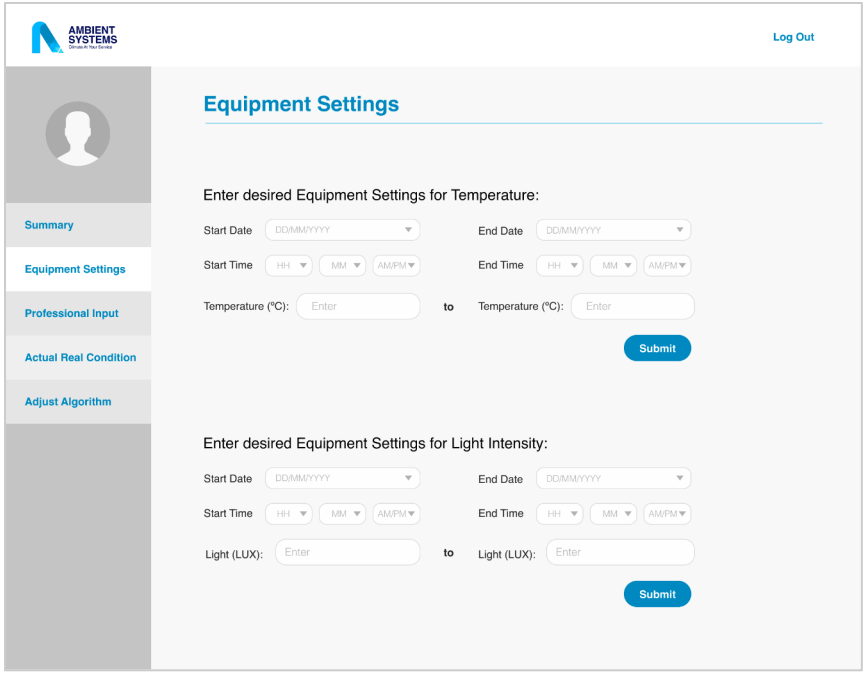
set of instructions from Ambient System's machine learning.

Dashboard

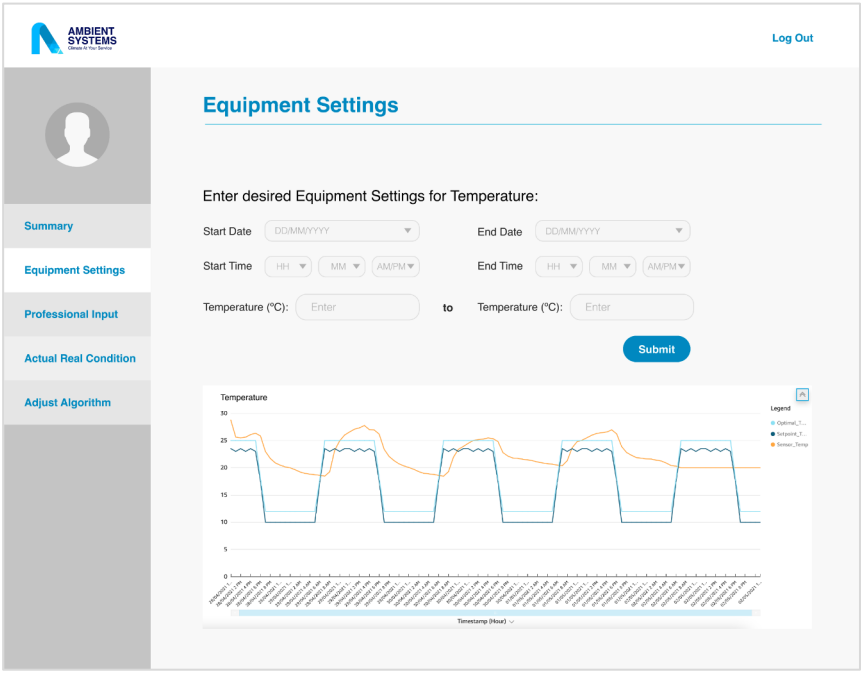
User flow



Dashboard (Data Visualization)



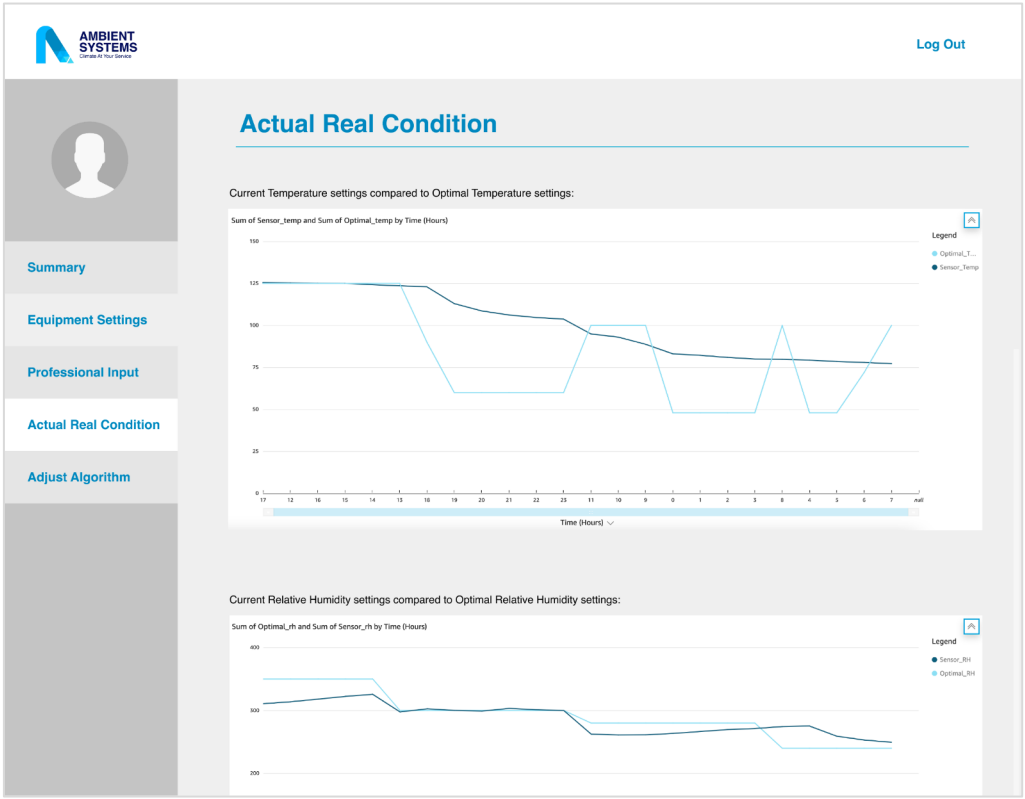
Before: User Inputs



After: Visualization Output

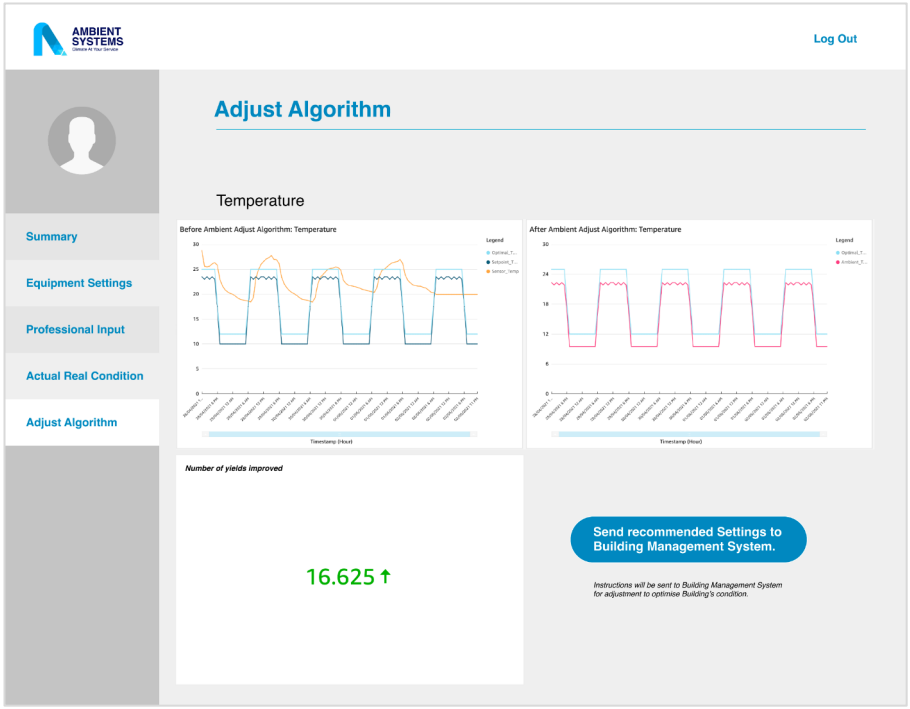
Dashboard (Data Visualization)

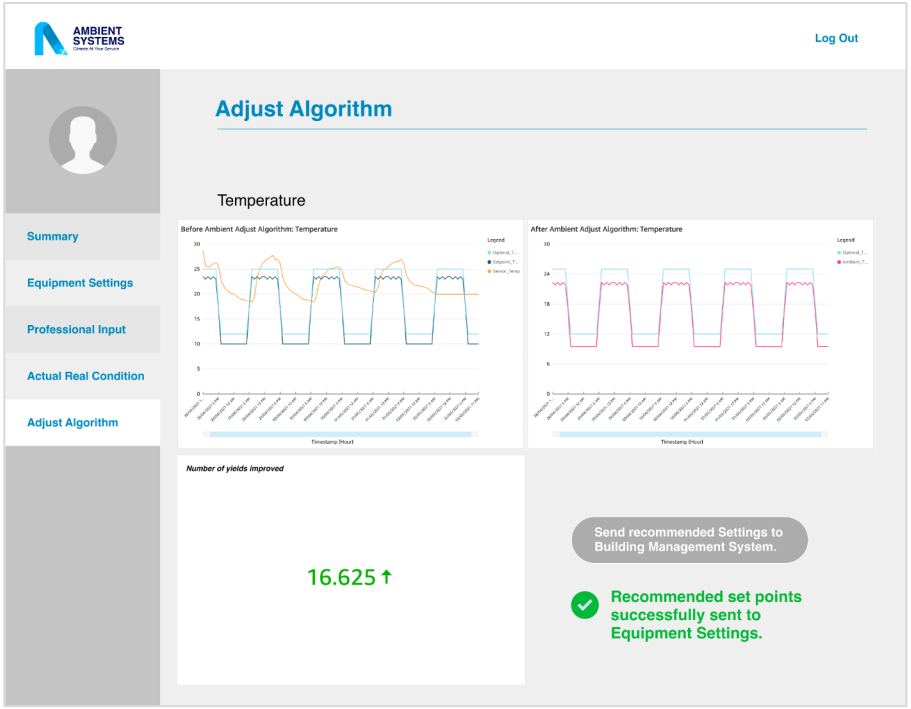
Datas from different sensors in the environment.



Dashboard (Data Visualization)

Recommended set of instructions from machine learning to enhance building performance.





Use Case Example

Simplified relationship between the current environment and Ambient Systems

If an Air Conditioner setting is set at 22 degree celsius, the room temperature might be at 25 degree celsius due to other factors affecting the room temperature.

Hence, Ambient Systems is able to detect the current room temperature, and recommend adjusted setpoint instructions for the Air Conditioner to lower the room's temperature to get desired condition.



Use Case Example

Agriculture Sector

1. Collecting data from different zone of the Building

• Displaying data detected in one Dashboard from sensors and equipment settings

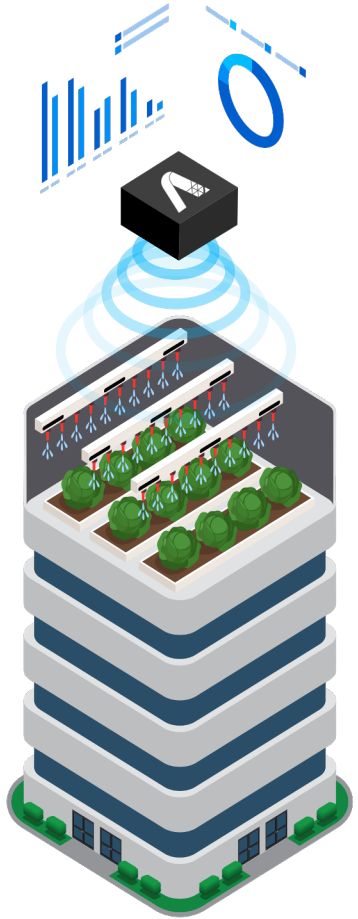
2. Analyses conditions affecting crop performance

• Machine learning evaluates any correlation among parameters, uncovering insights enhancing crop performance

3. Recommends a set of instructions to adjust equipment settings to achieve optimal crop performance

• Ambient Systems machine learning analyses the current data, compare the set of data to the ideal recipe (from step 2), and recommends user a set of instructions to adjust the equipment settings for desired environment (may be automated).

Parameters measured: Temperature, Relative Humidity, Air Quality, Airflow, Lighting, Soil Contents



Use Case Example

Healthcare Sector

- Ambient can be deployed in
  - Intensive and High Care Units
  - Outpatient Clinics
  - Patient wards
  - Operating Rooms
  - Pharmacies etc
  - Nursing Homes
- Temperature and RH controls are crucial for maintaining drug and vaccine safety, purity and efficacy.
- Air Quality control is crucial for maintaining hospital's staff or patients' health, to prevent spread of airborne disease, and also reduces risk of infectious diseases.
- Building management solution can monitor and control humidity levels and air changes in different zones of the hospital. (Zone Control Bacteria Detection)
- Room temperature, airflow, lighting can be adjusted to desired settings for times when the room is unoccupied, scale back energy use through reducing light and changing temperature settings.
- Able to integrate patient's data and clinical monitoring system?
- Increases hospital staff and facilities managers' productivity.

Parameters measured: Temperature, Relative Humidity, Air Quality, Airflow, Lighting

Use Case Example

Retail Sector

- Generate heat maps to indicate areas of heaviest foot traffic and dwell times in store to provide the data of customer navigation patterns, and to improve displays set up or re-distribute staff to locations with high traffic.
- Store managers can determine if low lighting has an impact on theft ratios in certain zones of the store.
- Monitor CO2 levels, ambient temperature and relative humidity in each zone throughout the day.
- Collect and analyze data from multiple sensors for actionables, to enable better use, optimize and prioritize building spaces while ensuring high levels of comfort for occupants.

Parameters measured: Temperature, Relative Humidity, Air Quality, Airflow, Lighting, Heat maps

Value Creation

After successfully updating equipment settings using Ambient's proprietary algorithm.

- Energy Savings, and reduced
- Man-hours Savings
- Reduction in Crop Loss (For Indoor Farms)
- Prevent spread of Airborne Disease (For Hospitals)

Applications



1. AgTech

Indoor farming sustainability & optimization



2. Healthcare

Patient well-being



3. Smart Homes

Cyber security of home-based IoT systems



4. Retail

Customer experience and inventory optimization

Thank You

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